Map Chat

Plan:

1. 4-vectors
2. Not 4-vectors (QM/QFT I)
3. Groups & Reps (a nice story about topology)
4. tensors & co-variant variables (as good as it gets)
5. Higher dim. games just for fun (quaternions & octonions?)

Pt. 1 | 4-vectors

Since spacetime sym.s are associated w/ Poincaré transformations realized as unitary* operators, we've learned to form Lorentz **invariant** quantities using Lorentz **covariant** variables.
This is the point of the "irreps of $SO(1,3)$"
The most commonly used is the $(\frac{1}{2},\frac{1}{2})$ vector rep. (on-shell cond. / Casimir inv.)

$$p^\mu = (p^0, \vec{p}), \quad p_i^2 = m_i^2$$

3-vectors like $\vec{p}$ aren't so useful b/c they're very frame-dep.
(n.b. by "frame dep" we mean, under spacetime sym. transformations, they transform like crap) p^μ is better, covariant!

For this reason, only talk about LI's!
So m always means m_0
All kinem. data: $\{m_i, s_{ij} := (p_i \pm p_j)^2\}$ ← basis! let's you ansatz
The presence of a sym. group constrained a lot! ↑

Pt. 2 ~~4-vectors~~

I'll now try to convince you that 4-vectors can be the wrong choice of irrep in very natural contexts, & that we should unlearn the physicists bias. Math knows better

Our context will be describing Physical states on Poincaré spacetimes, Particles interacting. N.b. the moral of our story will apply to any Problem w/ a Sym. group, not just Lorentz.

Let's then review the description of particles in QM/QFT! The problem w/ 4-vectors won't be the kinematics, it will be in the "Physical States"

Particle States: "Wigner's* method of induced rep.s"

Thm: There are no finite-dim. unitary rep.s of Poincaré

but, we can induce a stratification (a grading $\bigoplus_s V_s$) that sorts our ∞-dim. rep usefully.

And, we all already know how to do it! (mostly)

1) Boosts aren't unitary, so define a representative of the equiv. class (image) of momenta under LT's

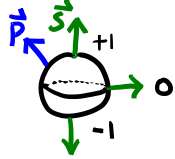
$$\left(K^{\pm}_{m^2\neq 0}\right)^\mu = \left(\pm m, \vec{0}\right), \quad \left(K^{\pm}_{m^2=0}\right)^\mu = \left(\pm 1, 0, 0, 1\right)$$

We can work over K^μ then boost to w/e you want

2) To encode spin d.o.f. recall from QM

$|s, j_{\hat{s}}\rangle$ has $2s+1$ states labelled by $J_{\hat{s}}$ eigenvalues, $j_{\hat{s}} \in [-s, s]$

$s=1$



$m^2 \neq 0$



$m^2 = 0$

Massless states only have 2 states, helicity states w/ $\vec{p} \parallel \vec{s}$ (Casimir inv.! Lorentz inv.!

Pauli-Lubanski pseudo-vector generates the lg, & encodes spin

$W^\mu W_\mu = m^2 s(s+1)$

but n.b. that $\vec{s}$ projections aren't frame-ind just like $\vec{p}$ wasn't! unless $m^2=0$? huh?

Tangent:

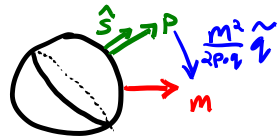
(*) **Thm:** Any on-shell massive momentum can be decomposed non-uniquely i.t.o 2 null momenta. $M \sim p + m^2 q$

$$\star \quad M^\mu = r^\mu + \alpha q^\mu \quad p^2 = 0 = q^2, \; p^\mu q_\mu \neq 0 \quad \star$$

Choose $r^\mu = p^\mu, (\vec{p} \parallel \vec{s}) \Rightarrow \alpha = \frac{m^2}{2p\cdot q}$ check!

Say we project p^μ out of q^μ so $\tilde{q}^\mu \perp \hat{s}^\mu$

then



So as $m^2 \to 0$, $M^\mu \to \pm p^\mu \parallel \hat{s}$

Re - Lorentz invariance

$\vec{S}$ isn't a LI b/c I can "boost past" the particle
So $\wedge p^\mu \to q^\mu$ & $\vec{S}\cdot\vec{p}$ gets all mixed up.
We're expanding our spin-dof in 3-vector space...
But, if the particle is massless it moves at c,
So I can't "boost past", & helicity is a LI

IE, $\hat{S}$ is a 3-vec. projection of Pauli-Lubanski,
$\pm h$ are the eigenvalues of the inv. operator
$\vec{w} \parallel \vec{p} \Rightarrow w^0 \sim -\vec{S}\cdot\vec{p}$ helicity operator

IEE, since the decomp. isn't unique for M^μ, of course
things $\sim \vec{m}$ aren't invariant. for p^μ the decomp.
is unique ($/\mathbb{Z}_2$)

3) Now we can split $|p,s,\cdots\rangle$ into "$|p\rangle \otimes |s,\cdots\rangle$" (for $s=1$, eg)
(for $s>1$ need $\varepsilon_s^{\mu\nu}$, etc)

$$\Psi_s^\mu(x) = \int \widetilde{dp}\, \varepsilon_s^\mu(p) e^{ipx}$$

Ψ_s^μ: a field; $\widetilde{dp}$: LI measure; $\varepsilon_s^\mu(p)$: Pol. vector.; e^{ipx}: Plane-wave exp

We're embedding a Lorentz rep. into a Lorentz vector field
— uh-oh

4) The Problem: Physical states transform under the $lG \subset LG$, but the lg & LG rep theory may not be the same! ε^μ has LG indices, but should have lG indices!

Pol. vectors transform like shit
Even worse, $2(1)+1 = 3$ states,
$\varepsilon^\mu \sim$ 4 dof
Impose gauge red. to fix mismatch in dof. For massless / $s>1$ it's even worse
$\vdots$
∞ traceless sym. $h^{\mu\nu}$

Illustration a la Dirac:

A massive spin-$\frac{1}{2}$ fermion

like massless ∞

$2(\frac{1}{2})+1 = 2$ dof

$$(i\not{\partial} - m)\psi = 0 \rightsquigarrow (i\gamma^\mu \partial_\mu - m)\psi = 0$$

$$\sim \begin{pmatrix} -m & i\sigma^\mu_{a\dot{a}}\partial_\mu \\ i\bar{\sigma}^\mu_{\dot{a}a}\partial_\mu & -m \end{pmatrix} \begin{pmatrix} \lambda^a \\ \rho_{\dot{a}} \end{pmatrix} = 0 \qquad \sigma^\mu = (\mathbb{1}_{2\times 2}, \vec{\sigma}) \quad a,\dot{a} = 1,2$$

$$\Downarrow$$

$$p_\mu (\sigma^{a\dot{a}})^\mu \rho_{\dot{a}} = m\lambda^a$$

$$p_\mu (\bar{\sigma}_{a\dot{a}})^\mu \lambda^a = m\rho_{\dot{a}}$$

$p^{a\dot{a}} := p^\mu (\sigma^{a\dot{a}})_\mu$ (2×2)

fermion & anti-fermion are each 2-comp spinors related by the dirac eqn. $\psi_D{}^\mu$ combines them, ew

$$p^{a\dot{a}}\rho_{\dot{a}} = m\lambda^a$$
$$p_{a\dot{a}}\lambda^a = m\rho_{\dot{a}}$$

Not only is the eqn more clear, the LG acts as $SO(2)$, natural for spinorial 2×2 $p^{a\dot{a}}$

So why bury it in a ψ?
work i.t.o the natural variables (here, spinors).

Nb. historically, spinor-notation was used for dumb reasons.
Later we understood better why $m^2=0$ is so special.
Now we can start from $m^2\neq 0$, understand all the structure, & explain the "special" massless behaviour as $m^2 \to 0$

Pt. 3 | Groups & Reps

Consider the LG in d-dim, $SO(1,d-1)$
$SO(d) \sim$ analytic cont.

$$\mathfrak{lG} \sim \begin{cases} ISO(d-1) \\ ISO(d-2) \end{cases}, \quad \begin{matrix} m^2 \neq 0 \\ m^2 = 0 \end{matrix}$$

We can largely ignore the I, which generates gauge trans. in fields. Since we'll work directly in $\mathfrak{lG}$ variables we won't have any.

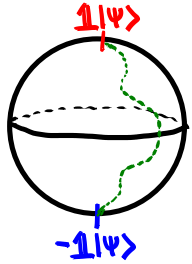
$d=4$ $\quad LG \sim SO(4) \sim SU(2)_L \otimes SU(2)_R$
$\sim SU(2)/\mathbb{Z}_2$ — "double" cover
$\downarrow$
$SO(1,3) \sim SL(2,\mathbb{C})/\mathbb{Z}_2$ — "double" cover

topology of $SO(1,3) \sim$ (Riemannian manifold) $\mathbb{R}^4 \times \mathbb{R}^3 \times (S^3/\mathbb{Z}_2)^{000}$

$e^{i\theta \cdot S} \sim e^{i\frac{\vec{\theta}\cdot\vec{\sigma}}{2}}$, $\frac{\theta}{2}$!

$\mathbb{R}^4$: boosts; $\mathbb{R}^3$: translations; $S^3/\mathbb{Z}_2$: rots/2

Not simply connected! $S^3/\mathbb{Z}_2$ has big consequence!
↳ antipodal points on S^3 ID'd
$|\psi\rangle = -|\psi\rangle$

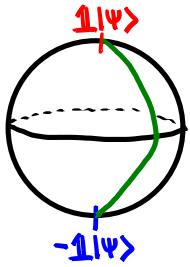


Take an int. rot $\mathbb{1} + i\vec{\theta}\cdot\vec{S}$, $[S^i, S^j] = \epsilon^{ijk} S^k$, $= \frac{1}{2}\vec{\sigma}$

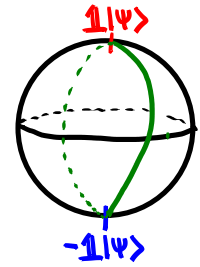
Exponentiating $\rightsquigarrow$ "walk" a path around S^3

Eg) $e^{2\pi i S_z}|\psi\rangle = -|\psi\rangle$ ($\underline{\text{not}}$ contractable!)

$e^{2\pi i S_z} e^{2\pi i S_z}|\psi\rangle = +|\psi\rangle$ (contractable!)



Non-Contractable



Contractable

The final state should be inv. under deformations of the "path", so contractable vs. non-cont.able are $\underline{\text{fundamentally different}}$

IE, since our topology is not simply-connected, our manifold has non-trivial 1st homotopy group

$$\Pi_1(S_3/\mathbb{Z}_2) = \mathbb{Z}_2$$

What is the $\underline{\text{meaning}}$ of this?
Consider rotating by 2π & 4π, from $|+s\rangle \to -|+s\rangle$ & $|+s\rangle \to |+s\rangle$,

$e^{2\pi i S}|\psi\rangle = -|\psi\rangle$

$e^{4\pi i S}|\psi\rangle = +|\psi\rangle$

$\Rightarrow$

$2\pi i S = i\pi + 2\pi i n$

$4\pi i S = 2\pi i n$ ($n = 0, 1, \ldots$)

(true for $S = \frac{1}{2} + n$, also for $S = n$)

fermions: $S_{nc} = \frac{1}{2} + n$

bosons: $S_c = 1 + n$

Spin-statistics is a feature of the $\underline{\text{topology}}$ of the LG!

↓

a feature of the dbl cover

Some words of summary:

- $SU(2)$ double covers $SO(3)$
- doubling $SU(2) \mapsto SO(3)$
  - 2 spinors can replace a vector ← lowest-weight irrep of $SU(2)$
  - "Spinors are the $\sqrt{\ }$ of a vector"

  $D(\frac{1}{2},\frac{1}{2}) \sim D(\frac{1}{2},0) \otimes D(0,\frac{1}{2}) \sim SU(2)_L \otimes SU(2)_R$
- $SU(2)$ works as a natural object for both $SU(2)$ & $SO(3)$, & even $SO(2) \sim U(1)$

We should consider using $SU(2)$-objects b/c they are "natural" candidates & cover all mass & spin cases.

Pt. 4 Something New ~17

How to rep. Spin-States Redux:

Obviously labelling by j_z is dumb & not in line w/ the "covariant dogma".

Instead, use

Rank 2S symmetric $SU(2)$ tensors

$X^{\{I_1 \cdots I_{2S}\}} := X^{I_1 \cdots I_{2S}} + (\text{Sym.})$ $\quad I_i = 1,2$

Consider $X^{\{I\}} = \xi^{(I_1} \cdots \xi^{I_{2S})}$,

Expand each ξ^I into a 2-dim SU(2) basis
For our basis, pick the $\pm\frac{1}{2}$ helicity eigenvectors

$$\xi^I = \alpha\, \xi^{+I} + \beta\, \xi^{-I}$$

EG: Structure of $s=1$

$$\xi^{I_1}\xi^{I_2} = \underbrace{\alpha^2 \xi^+\xi^+}_{h=+1} + \underbrace{\beta\alpha\, \xi^-\xi^+ + \alpha\beta\, \xi^+\xi^-}_{j_\xi=0} + \underbrace{\beta^2 \xi^-\xi^-}_{h=-1}$$

$M^{j_s} \sim \left((\xi_i^+)^{s+j_s}(\xi_i^-)^{s-j_s}\right)$

Non-helicity terms $\sim O(m)$ just like q^μ

Each I index is an SU(2) index & transforms nicely under the LG (massive or massless, any spin!)

How to encode kinematics:

$P^{a\dot a} = (\sigma_\mu{}^{a\dot a})\, p^\mu$ ← 2x2 momentum-matrix

$\det(P) = m^2$, Fact 1: any 2x2 can be written as a lin. combo of 2 rank-1 (null) 2x2 mat.s

$\rightarrow P^{a\dot a} = K^{a\dot a} + m^2 q^{a\dot a}$

Fact 2: any rank 1 2x2 can be written lit.o. 2 2-vectors

$\rightarrow P^{a\dot a} = a|\lambda\rangle[\tilde\lambda|_{\dot a} + a|\lambda]\langle\tilde\lambda|_{\dot a}$

$(=)(--) + (=)(--)$

$K^{a\dot a} + q^{a\dot a}$

Again we can choose for our basis to expand s.t.
$|\lambda\rangle[\tilde{\lambda}|$ is the correct massless limit & $|\lambda]\langle\tilde{\lambda}| \sim O(m)$

Now for the magic to combine w/ the $\mathfrak{lG}$ $\mathfrak{S}^I$'s.
In the massless case we write $p^{a\dot{a}}$ in the helicity basis & recognize $|\lambda\rangle \sim$ (−) helicity
$|\tilde{\lambda}] \sim$ (+) helicity
What to make of $|\lambda]$ & $|\tilde{\lambda}\rangle$?
In this basis they're $O(m)$ & $|\lambda] \sim$ (+) helicity
$|\tilde{\lambda}\rangle \sim$ (−) helicity

So define

Dirac eqn

$$|\underline{\lambda}\rangle^I = |\lambda\rangle\mathfrak{S}^{-I} + |\lambda]\mathfrak{S}^{+I} \qquad P|\underline{\lambda}\rangle = m|\underline{\tilde{\lambda}}]$$

$$|\underline{\tilde{\lambda}}]^I = |\tilde{\lambda}]\mathfrak{S}^{+I} + |\tilde{\lambda}\rangle\mathfrak{S}^{-I}, \qquad P|\underline{\tilde{\lambda}}] = m|\underline{\lambda}\rangle$$

$$\begin{pmatrix} \underline{a} & \underline{c} \\ \underline{b} & \underline{d} \end{pmatrix} = \begin{pmatrix} \underline{a} \\ \underline{b} \end{pmatrix}(10) + \begin{pmatrix} \underline{c} \\ \underline{d} \end{pmatrix}(01)$$

$$\begin{pmatrix} |\tilde{\lambda}]^1 & |\tilde{\lambda}\rangle^1 \\ |\tilde{\lambda}]^2 & |\tilde{\lambda}\rangle^2 \end{pmatrix} = \begin{pmatrix} |\tilde{\lambda}]^1 & 0 \\ |\tilde{\lambda}]^2 & 0 \end{pmatrix} + \begin{pmatrix} 0 & |\tilde{\lambda}\rangle^1 \\ 0 & |\tilde{\lambda}\rangle^2 \end{pmatrix}$$

In this notation we can write

$$P^{a\dot{a}} = |\underline{\lambda}\rangle^I{}_I[\underline{\tilde{\lambda}}| = |\lambda\rangle[\tilde{\lambda}| + |\lambda]\langle\tilde{\lambda}|$$

LI's: $\{m_i^2, \langle\lambda\lambda\rangle, \langle\lambda\lambda], [\lambda,\lambda\rangle, [\lambda\lambda]$
$\langle\tilde{\lambda}\tilde{\lambda}\rangle, \langle\tilde{\lambda}\tilde{\lambda}], [\tilde{\lambda},\tilde{\lambda}\rangle, [\tilde{\lambda}\tilde{\lambda}]\}$

These variables suffice to describe massive particles w/ no gauge redundancy, & working directly i.t.o. $\mathfrak{lG}$ covariant variables to form LI's

In the massless limit $\rightsquigarrow$ mSH

Pt. 5 | Higher dim.s

The whole story hung on $su(2)$ being a good rep. for our sym. group in $d=4$. In fact other dim.s have similar stories!

$d=3$: $so(1,2) \sim sl(2,\mathbb{R})/\mathbb{Z}_2$

dbl cover by real valued 2-vectors
→ mSH ~ bi-vectors (real)

$d=4$: $so(1,3) \sim sl(2,\mathbb{C})/\mathbb{Z}_2$

dbl cover by complex 2-vectors
→ mSH ~ bi-spinors, MSH

$d=5$: $so(1,4) \sim usp(2,2)$
$lG \sim so(4)$ weird

~~dbl cover~~ by 4-cmpt vectors
→ mSH, MSH Radu & Murat!

$d=6$: $so(1,5) \sim su(4) \sim sl(2,\mathbb{H})/\mathbb{Z}_2$

dbl cover by unit quaternionic 2-vectors
→ mSH ~ bi-spinors, MSH in 4D
Hoos!?

$d=10$: $so(1,9) \sim su(8) \sim sl(2,\mathbb{O})/\mathbb{Z}_2$

dbl cover by unit octonionic 2-vectors
→ mSH ~ bi-spinors